1.3 Volumetric Analysis

1. Glucose can be converted to ascorbic acid (vitamin C), a white solid with a molar mass of 176 g mol^{-1}. Manufacturers of vitamin C tablets must analyse sample tablets from each production batch to maintain strict control of the quality of their product. Iodine is used as an oxidant in the analysis of ascorbic acid and reacts as shown in the following equation:

$$I_{2(aq)} + \underset{\text{ascorbic acid}}{C_6H_8O_{6(aq)}} \rightarrow 2I^-_{(aq)} + 2H^+_{(aq)} + C_6H_6O_{6(aq)}$$

Two tablets containing ascorbic acid, and having a total mass of 0.500 g, were crushed and dissolved in a small quantity of water then accurately diluted to 100 mL in a volumetric flask.

Three 25.0 mL portions of this solution were transferred to conical flasks and titrated with 0.0250 mol L^{-1} iodine solution from a burette. The average titre was 14.25 mL.

(a) Calculate the molar concentration of the ascorbic acid solution in the volumetric flask.

(4 marks)

(b) Calculate the mass of ascorbic acid in each tablet and hence the percentage of ascorbic acid in each tablet.

(3 marks)

Starch is added to the conical flasks to act as an indicator in these titrations. It turns blue-black in the presence of iodine.

(c) State the colour change that would indicate the end-point of the titration. (2 marks)

2. In order to carry out a titration the iodine solution to be used has to be standardised against a sodium thiosulfate solution of known concentration.

List the steps you would take to make up a 0.0250 mol L^{-1} standard sodium thiosulfate solution in a 250 mL volumetric flask when provided with the required mass of sodium thiosulfate (1.55 g) in a small beaker. Each step in your answer should include sufficient detail to ensure the accuracy of the stated concentration.

(7 marks)

3. The labels on commercial bleaches frequently state the concentration of sodium hypochlorite present.

SODIUM HYPOCHLORITE (NaClO)
concentrated liquid bleach for use in laundries or food processing areas
Contains: 12.5% available chlorine

The amount of available chlorine in a commercial bleach solution was determined by volumetric analysis. 20.0 mL samples of diluted bleach solution were pipetted into conical flasks to which excess potassium iodide and dilute sulfuric acid were added. The reaction of hypochlorite ions is shown by the equation below:

$$ClO^-_{(aq)} + 2H^+_{(aq)} + 2I^-_{(aq)} \rightarrow Cl^-_{(aq)} + H_2O_{(l)} + I_{2(aq)}$$

The iodine produced by the reaction in each flask was determined by titrating with 0.100 mol L^{-1} sodium thiosulfate solution from a burette, using starch as the indicator. The average titre was 12.75 mL.

The reaction that occurred during the titration is shown by the equation below:

$$2S_2O_3^{2-}{}_{(aq)} + I_{2(aq)} \rightarrow S_4O_6^{2-}{}_{(aq)} + 2I^-_{(aq)}$$

(a) Calculate the number of moles of thiosulfate in the average titre. (1 mark)

(b) Calculate the number of moles of iodine released by the bleach. (1 mark)

(c) Hence give the number of moles of hypochlorite ions in each 20.0 mL aliquot of diluted bleach. (1 mark)

(d) Calculate the concentration, in g L^{-1}, of sodium hypochlorite in the diluted bleach. (2 marks)

In order to make up the diluted bleach solution for the titration a 5.00 mL sample of the commercial bleach was pipetted into a 250 mL volumetric flask and made up to the engraved mark with distilled water.

(e) Calculate the concentration, in g L^{-1} and in grams per 100 mL, of sodium hypochlorite in the undiluted bleach. (2 marks)

(f) '12.5% available chlorine', as stated on the label, means that the concentration of sodium hypochlorite in the undiluted bleach should be 13.1 g per 100 mL.

Similar analyses consistently gave the concentration a lower value than that stated on the label. Suggest a reason for the lower value. (1 mark)

(g) Explain how each of the following procedures would have affected the titre values in this analysis.

(i) The pipette used to transfer the commercial bleach to the volumetric flask was rinsed with water immediately before it was filled with the bleach. (2 marks)

(ii) After the distilled water had been added to the volumetric flask the bottom of the meniscus was left below the engraved mark. (2 marks)

4. Effluent discharged from the Solvay process may contain ammonia. Samples of the effluent are therefore collected and analysed to determine the ammonia content. The analysis is carried out as follows:

Step 1: A 100.0 mL sample of effluent is boiled to drive off the dissolved ammonia.

Step 2: The ammonia driven off is bubbled into 10.0 mL of 0.0100 mol L^{-1} hydrochloric acid.

$$NH_{3(g)} + HCl_{(aq)} \rightarrow NH_4Cl_{(aq)}$$

Step 3: The unreacted hydrochloric acid is titrated with 0.0100 mol L^{-1} sodium hydroxide.

A titre value obtained in such a titration is 5.23 mL.

$$HCl_{(aq)} + NaOH_{(aq)} \rightarrow NaCl_{(aq)} + H_2O_{(l)}$$

(a) Calculate the number of moles of hydrochloric acid originally present in the 10.0 mL aliquot of hydrochloric acid. (2 marks)

(b) Calculate the number of moles of hydrochloric acid that reacted with the sodium hydroxide. This is the amount of unreacted hydrochloric acid remaining after the ammonia was bubbled into it. (3 marks)

(c) Calculate the number of moles of hydrochloric acid that reacted with the ammonia bubbled into it from the sample of effluent. (2 marks)

(d) Calculate the concentration of ammonia in the sample of effluent.

(i) in mol L^{-1} (3 marks)

(ii) in grams per 1000 L (that is, parts per million, ppm). (3 marks)

(e) A laboratory technician who carries out this analysis uses the following formula to calculate the concentration of ammonia from the sample of effluent in ppm:

(mL of 0.0100 mol L^{-1} $HCl_{(aq)}$ - mL of 0.0100 mol L^{-1} $NaOH_{(aq)}$) × 1.703

Use this formula to calculate the concentration of ammonia in ppm. (2 marks)

(f) Predict how the sodium hydroxide titre at Step 3 will change if the amount of ammonia in the effluent increases. Explain your prediction. (3 marks)

5. Chlorine bleaches commonly contain sodium hypochlorite (NaOCl) solution. One sample of household bleach contains 5.25 g of NaOCl per 100.0 mL.

(a) Calculate the concentration of this solution in mol L^{-1}. (2 marks)

(b) Calculate the number of moles of NaOCl in 1000.0 L of such a solution. (1 mark)

(c) Sodium hypochlorite solution may be prepared by bubbling chlorine into a sodium hydroxide solution, as shown below:

$Cl_{2(g)} + 2NaOH_{(aq)} \rightarrow NaCl_{(aq)} + NaOCl_{(aq)} + H_2O_{(l)}$

Calculate the mass of chlorine required to prepare 1000.0 L of this solution. (2 marks)

6. The alcohol content of a wine was determined by the following procedure (Steps 1 to 4):

Step 1: The ethanol was distilled from a 100.0 mL sample of the wine. The ethanol distillate was then diluted to 100.0 mL.

(a) Suggest why the distillate was diluted to 100.0 mL. (1 mark)

Step 2: A 10.0 mL sample of this ethanol solution was added to a 20.0 mL aliquot of potassium dichromate solution.

(b) Calculate the number of moles of potassium dichromate originally in the 20.0 mL aliquot of 1.25 mol L^{-1} potassium dichromate solution. (2 marks)

Step 3: The mixture was then heated at 60°C for 30 minutes for oxidation to take place, as shown in the following equation:

$2Cr_2O_7^{2-} + 3CH_3CH_2OH + 16H^+ \rightarrow 3CH_3COOH + 4Cr^{3+} + 11H_2O$

Step 4: The unreacted potassium dichromate was titrated with an iron (II) solution, as shown in the following equation:

$Cr_2O_7^{2-} + 6Fe^{2+} + \_\_\_H^+ \rightarrow 2Cr^{3+} + 6Fe^{3+} + \_\_\_H_2O$

(c) Balance the equation shown in Step 4. (1 mark)

(d) The amount of potassium dichromate left after reaction with the ethanol was 6.91×10^{-3} mol.

(i) Calculate the volume of 2.00 mol L^{-1} iron(II) solution required to react completely with the remaining potassium dichromate. (3 marks)

(ii) Calculate the number of moles of potassium dichromate that reacted with the ethanol in the 10.0 mL sample. (1 mark)

(iii) Show that there were 0.0271 mol of ethanol in the 10.0 mL sample. (2 marks)

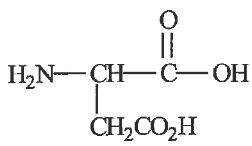
(iv) Calculate the concentration of ethanol in the 10.0 mL sample in mol L^{-1} (1 mark)

(v) Calculate the mass of ethanol in the 100.0 mL sample of wine. (2 marks)

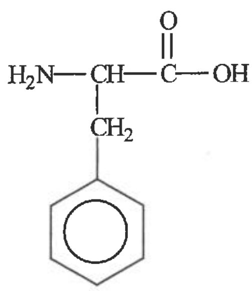
(vi) Explain whether the volume of iron(II) solution needed in Step 4 would have been increased, decreased, or unchanged if the mixture had not been heated in Step 3.
(3 marks)

1.4 Chromatography

1. Explain, using the principles of adsorption chromatography, why compounds A and B (shown below) can be separated using a relatively non-polar solvent compared to the thin layer plate.
(5 marks)

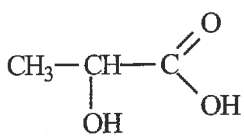


A

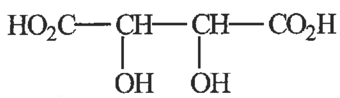


B

2. Lactic acid (**B**) and tartaric acid (**C**), whose formulae are shown below, are present in small amounts in red wines and contribute to the characteristic taste of the wine.



B



C

Thin layer chromatography can be used to separate mixtures containing compounds such as **B** and **C** because of their selective binding to the plate. A sample of red wine containing both lactic acid and tartaric acid was separated by thin layer chromatography, using an appropriate solvent.

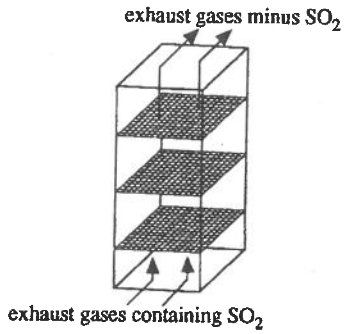
(a) Explain which of the two molecules is the more polar (2 marks)

(b) State, giving a reason for your answer, which compound, **B** or **C**, would be adsorbed more strongly onto the plate using aluminium oxide as a substrate during chromatography. (2 marks)

Topic 2: Managing Chemical Processes

2.1 Rates of Reactions

1. Sulfur dioxide is a component of exhaust gases released into the atmosphere by several industrial processes. Because sulfur dioxide is now known to be a major cause of acid rain, chemists are devising ways in which to remove it from exhaust gases before their release into the atmosphere.

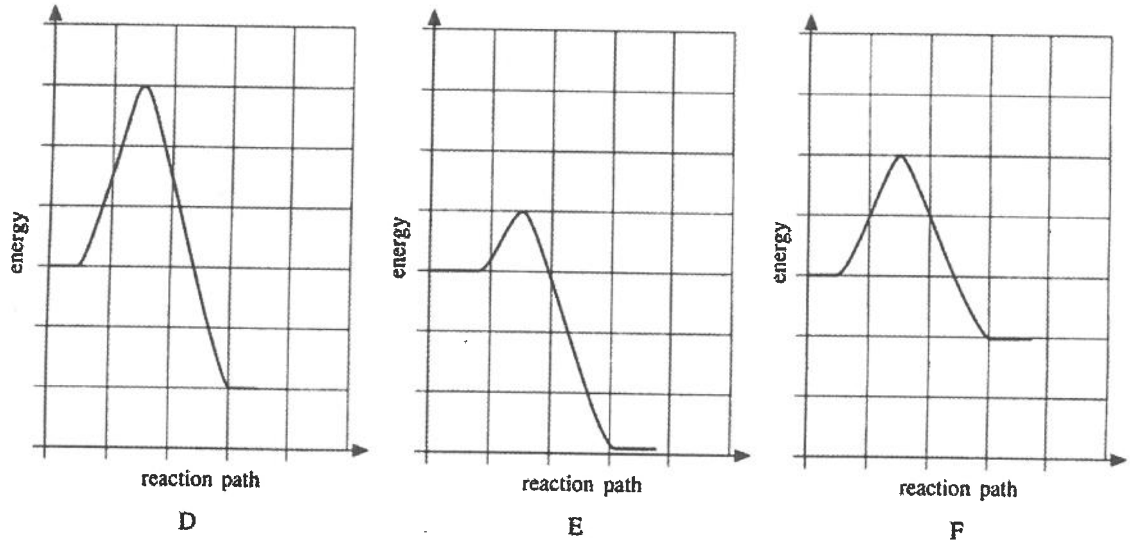


In one such process, known as 'dry scrubbing', exhaust gases are passed over powdered lime (calcium oxide, CaO) coated onto a fine wire gauze. The sulfur dioxide is converted to calcium sulfite, as shown by the following equation:

$$CaO_{(s)} + SO_{2(g)} \rightarrow CaSO_{3(s)}$$

Explain why the lime is powdered and coated onto a fine wire gauze. (3 marks)

2. The diagram below shows the energy profile graphs for the separate combustion of three liquid fuels, coded D, E, and F:

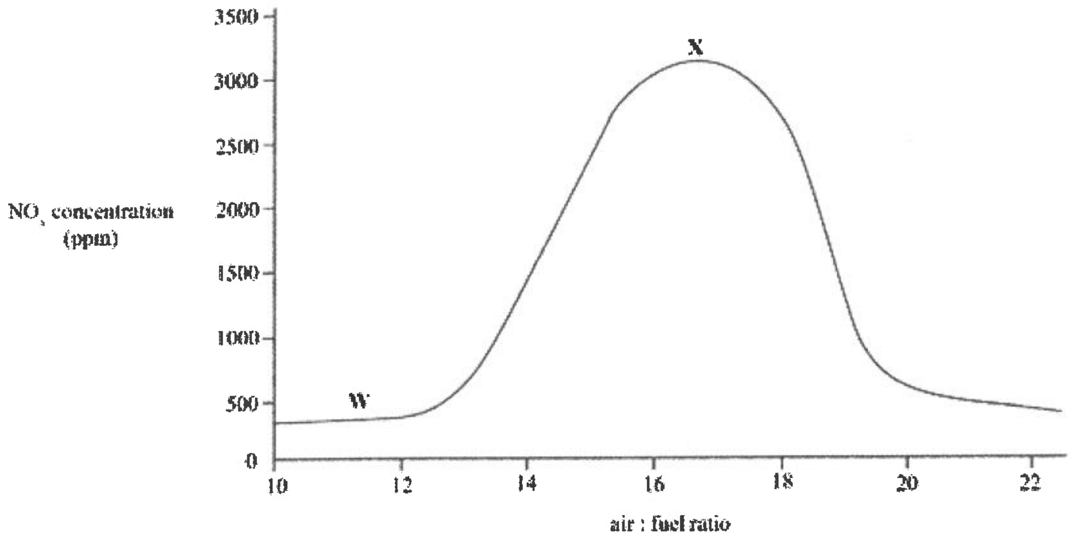


(a) Predict, from the information provided, which fuel has the highest activation energy. Give a reason for your answer. (2 marks)

(b) (i) Write the code letter of the liquid fuel that has the greatest enthalpy of combustion. (1 mark)

(ii) Indicate on the graph the magnitude of its enthalpy of combustion. (1 mark)

3. The combustion of petrol produces a number of emissions, including nitrogen oxides (NO_x). The proportions of these products vary with the amount of air used in the engine.



(a) Using the graph, determine the concentration of NO_x (in ppm) produced with an air : fuel ratio of 15. (2 marks)

(b) State one reaction condition necessary for direct reaction between nitrogen and oxygen. (1 mark)

(c) Explain why the greater amount of air available at X than at W leads to the production of more NO_x. (2 marks)

(d) Suggest why an engine is not usually run with a low air : fuel ratio, which would produce less NO_x. (2 marks)

4. Hydrogen peroxide solution can be used as a bleaching agent. It decomposes as shown by the equation below:

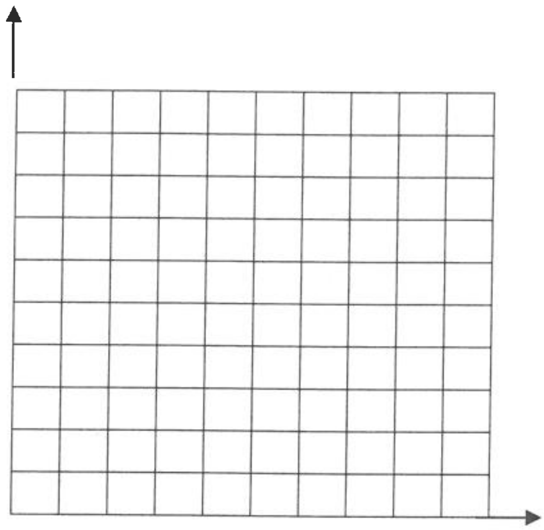
$$2H_2O_{2(aq)} \rightarrow 2H_2O_{(l)} + O_{2(g)}$$

An experiment carried out to monitor the decomposition of hydrogen peroxide gave the following data for the concentration of hydrogen peroxide remaining at the end of each time interval shown.

$[H_2O_2]$ mol L^{-1}	Time (minutes)
2.32	0
1.86	5
1.49	10
0.98	20
0.62	30
0.25	50

(a) On the axes on the following page, plot the concentration of hydrogen peroxide remaining against the time at which the concentration was determined.

Graph of [H_2O_2] remaining against time



(SIS) (6 marks)

(b) On the axes above, sketch the following two graphs:

(i) The graph expected if the experiment had been carried out at lower temperature. Label this graph P. (1 mark)

(ii) The graph expected if a catalyst had been added before the experiment was started. Label this graph Q. (1 mark)

(c) Use the collision theory to explain why the concentration of hydrogen peroxide fell more rapidly in the first 5 minutes than it did in the 5-minute interval between 25 minutes and 30 minutes. (3 marks)

2.2 Equilibrium and Yield

1. Fuel-grade ethanol can be produced synthetically by a reaction between ethene and steam. The reaction is carried out at 300 °C and 60 atmospheres pressure, in the presence of a catalyst, as shown by the equation below:

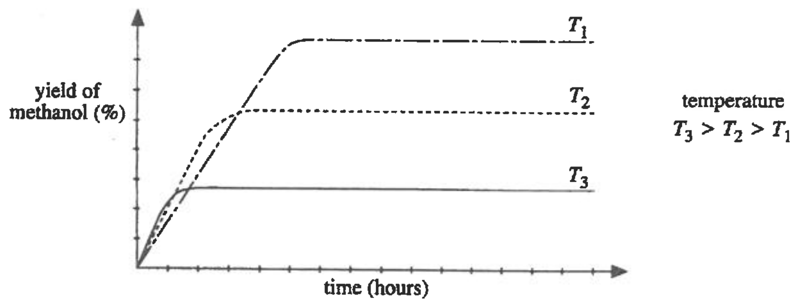
$$C_2H_{4(g)} + H_2O_{(g)} \rightleftharpoons C_2H_5OH_{(g)} \qquad \Delta H = -46 \text{ kJ mol}^{-1}$$

(a) Explain the effect of the high pressure on the yield of ethanol produced by this reaction. (3 marks)

(b) Methanol, another synthetic fuel, is produced industrially by a reaction between carbon monoxide and hydrogen, as shown by the equation below:

$$CO_{(g)} + 2H_{2(g)} \rightleftharpoons CH_3OH_{(g)}$$

The graph below shows the effect of temperature on the yield of methanol produced by this reaction:



(1) Write the expression for K_c for the reaction given. (1 mark)

(2) State what the graph above shows about the yield of methanol as the temperature is increased. (1 mark)

(3) Using your answer to part (2), and giving reasons, state whether the production of methanol is endothermic or exothermic. (3 marks)

(4) T_2 is the preferred operating temperature for this industrial process. Suggest why T_2 is chosen in preference to either T_1 or T_3. (2 marks)

(2013, Q5)

2. Sulfur dioxide is added to white wines as an antioxidant and bactericide. It reacts with water to give the hydrogensulfite anion, as shown by the equations below:

$$SO_{2(aq)} + H_2O \rightleftharpoons H_2SO_{3(aq)}$$

$$H_2SO_{3(aq)} \rightleftharpoons H^+_{(aq)} + HSO_3^-{}_{(aq)}$$

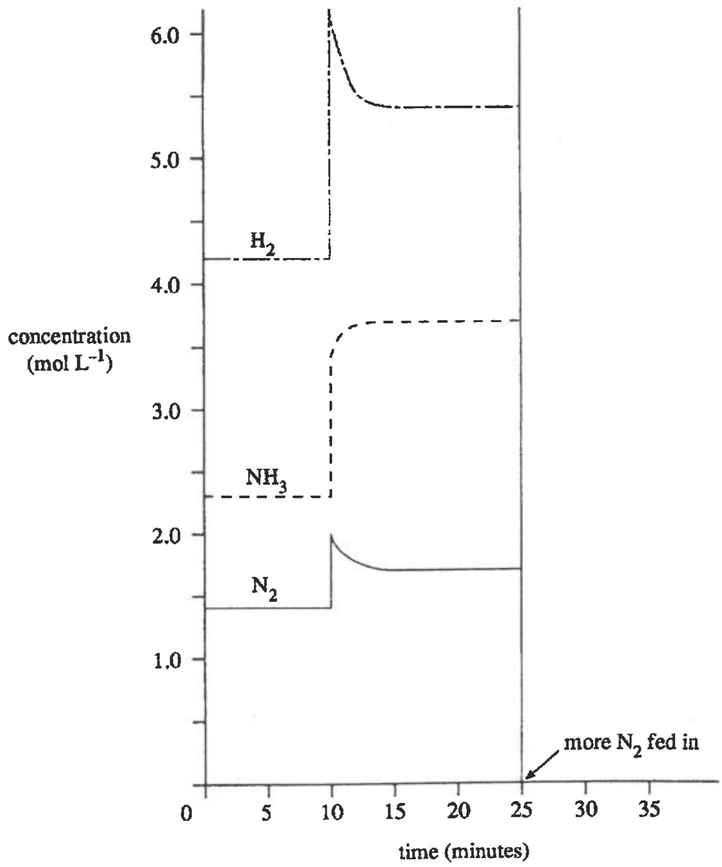
Explain why the amount of sulfur dioxide decreases as the pH of the wine is increased.(4marks)

3. A study of the equilibrium between N_2, H_2, and NH_3 gases during the production of ammonia was carried out in a closed reaction vessel connected to supplies of N_2 and H_2. The temperature was maintained at a constant value.

The equation for the reaction occurring inside the vessel is:

$$N_{2(g)} + 3H_{2(g)} \rightleftharpoons 2NH_{3(g)}$$

Monitoring equipment continuously measured the concentrations of the N_2, H_2, and NH_3. The changes in concentration were recorded on a chart, part of which is shown below. Use the chart to answer questions (a) to (d) on the next page.



(a) Write the equilibrium expression K_c for the reaction that occurred in the vessel and calculate its value, using the concentrations of each gas during the first 10 minutes. (5 marks)

(b) Suggest the change that occurred at the 10-minute mark and explain the consequent changes that occurred in the reaction mixture during the following 2 minutes. (4 marks)

(c) State how you recognised that equilibrium had been re-established at the 15-minute mark. (1 mark)

(d) At the 25-minute mark, more nitrogen was fed into the reaction vessel. Draw extensions on the graph to show the effect of the increased amount of nitrogen on the concentrations of each of the gases, N_2, H_2, and NH_3, during the following 10 minutes. (4 marks)

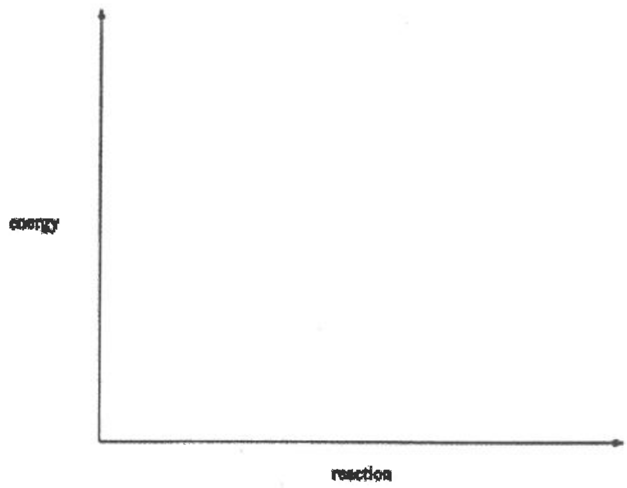
4. Exhaust pollutants can be passed through catalytic converters that remove carbon monoxide, nitrogen dioxide, and hydrocarbons by redox processes.

(a) Nitrogen dioxide may be converted into nitrogen by reduction with carbon monoxide. Write a balanced equation for this reaction. (2 marks)

(b) Suggest why, in a catalytic converter, the catalyst is supported on a porous material. (1 mark)

(c) Write the equation for the reaction taking place in the catalytic converter which reduces the nitric oxide concentration. (2 marks)

(d) Complete the energy profile diagram below to show how a catalyst affects the exothermic reaction between nitrogen dioxide and carbon monoxide. (3 marks)



(e) Nitric oxide (NO) is formed at high temperatures in internal-combustion engines, according to the following equation:

$$N_{2(g)} + O_{2(g)} \rightleftharpoons 2NO_{(g)} \quad \Delta H = +181 \text{ kJ mol}^{-1}$$

(i) State the effect of the high temperature on the rate of the reaction. (1 mark)

(ii) Write the equilibrium expression for this reaction. (1 mark)

(iii) Calculate the value of K_c if an equilibrium mixture contains 0.035 mol L^{-1} of nitrogen, 8.9×10^{-3} mol L^{-1} of oxygen, and 1.4×10^{-3} mol L^{-1} of nitric oxide.(2 marks)

(iv) Suggest why exhaust gases from internal-combustion engines are not equilibrium mixtures. (1 mark)

5. A problem associated with the use of fossil fuels is the photochemical smog produced by motor vehicle emissions. Nitrogen dioxide is a key component of photochemical smog. It is produced by oxidation of nitric oxide, as shown in the following equation:

$$2NO_{(g)} + O_{2(g)} \rightleftharpoons 2NO_{2(g)}$$

At a temperature of 1500 K, the equilibrium constant, K_c, for this reaction is 1.6.

(a) Write the K_c expression for this reaction. (2 marks)

(b) In a laboratory sample, taken at 1500 K, the following mixture of gases was obtained:

0.050 mol L^{-1} nitric oxide 0.080 mol L^{-1} oxygen 0.010 mol L^{-1} nitrogen dioxide

(i) Determine whether or not this mixture had reached equilibrium. Clearly show your working. (3 marks)

(ii) At 2000 K, the equilibrium constant is lower than at 1500 K. By using this information, determine whether this reaction is exothermic or endothermic. (3 marks)

6. Some hydrogen iodide gas was placed in a closed reaction vessel of volume 1 L. The temperature was maintained at 445°C until equilibrium was established 10 minutes later. The reaction that occurred in the vessel is shown in the following equation:

$$\underset{\text{colourless}}{2HI_{(g)}} \rightleftharpoons H_{2(g)} + \underset{\text{purple}}{I_{2(g)}} \qquad \Delta H = +10\text{ kJ}$$

The initial and equilibrium concentrations of each component are given in the table below:

Component	**Initial Concentration (mol L^{-1})**	**Equilibrium Concentration (445° C) (mol L^{-1})**
HI	x	43×10^{-4}
H_2	0	6.0×10^{-4}
I_2	0	6.0×10^{-4}

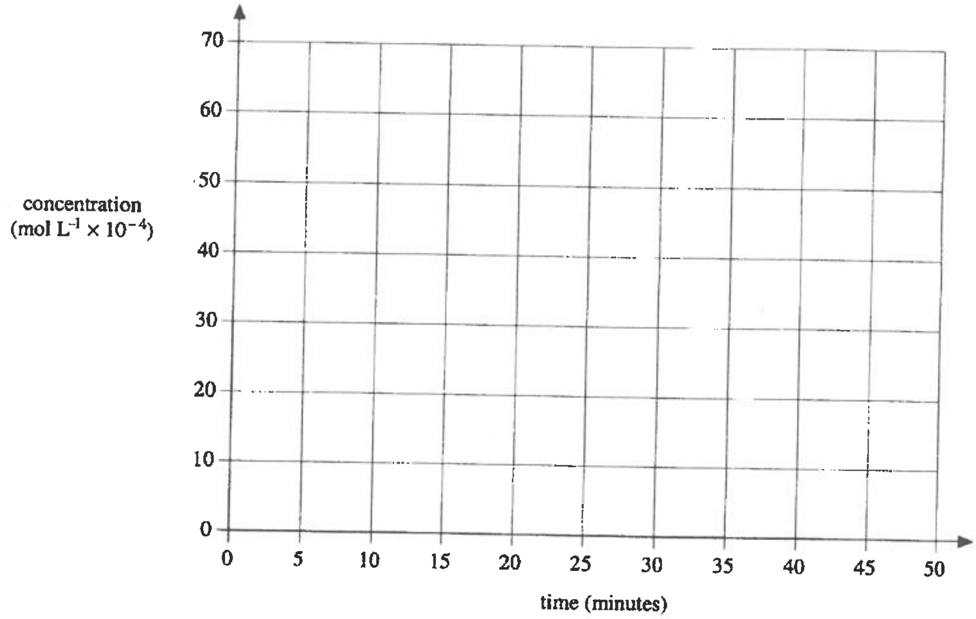
(a) Calculate the initial concentration, x, of the hydrogen iodide. (2 marks)

(b) State the observation that would enable you to recognise that equilibrium had been established. (1 mark)

(c) Write the expression for K_c for this reaction. (1 mark)

(d) Calculate the value of K_c for the decomposition of hydrogen iodide at 445° C. (2 marks)

(e) On the axes below plot the graphs of the concentrations of the hydrogen iodide and the iodine from the beginning of the experiment until 10 minutes after the establishment of equilibrium at 445° C. (4 marks)

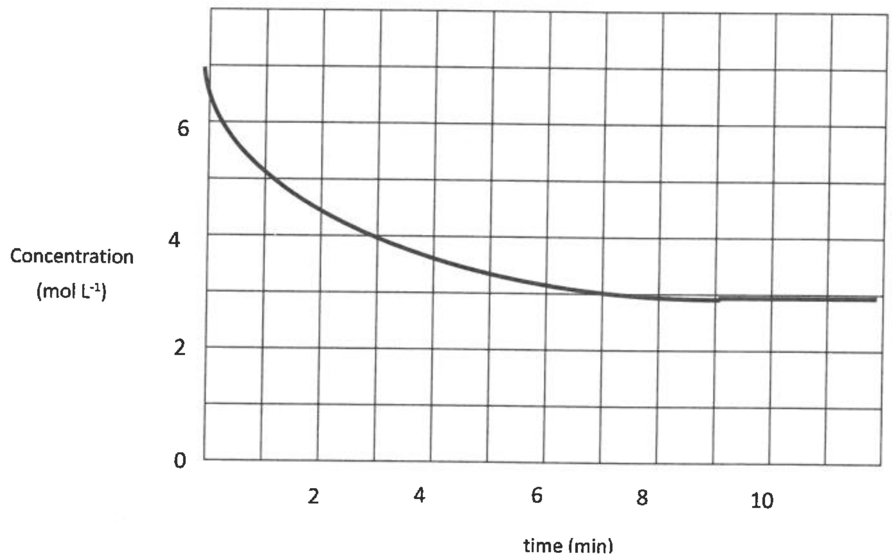


(f) Extend your graphs to show what you would expect to happen to the concentrations of the hydrogen iodide and the iodine if the reaction mixture at equilibrium at 445°C was cooled to 395°C and again allowed to reach equilibrium. (3 marks)

7. The equilibrium reaction that occurs between nickel and carbon monoxide is shown in the equation below:

$$Ni_{(s)} + 4CO_{(g)} \rightleftharpoons Ni(CO)_{4(g)}$$

The graph below shows how the concentration of carbon monoxide changes as a mixture of nickel and carbon monoxide comes to equilibrium.



(a) State the time at which equilibrium is reached. (1 mark)

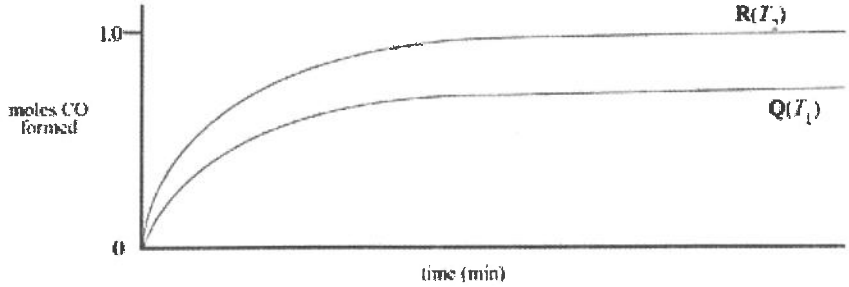
(b) Given that there was no $Ni(CO)_{4(g)}$ present originally, calculate its concentration at equilibrium. (2 marks)

(c) On the axes above, draw a graph to show how the concentration of $Ni(CO)_{4(g)}$ changes from time = 0 until time = 11 minutes. (3 marks)

8. Methane is commonly used to prepare hydrogen for industrial purposes by heating natural gas with steam in the presence of a catalyst, as shown below:

$$CH_{4(g)} + H_2O_{(g)} \rightleftharpoons 3H_{2(g)} + CO_{(g)}$$

When 1 mole of methane was reacted with excess steam at temperature T_1, curve Q in the following diagram was obtained. When the experiment was repeated at temperature T_2, curve R was obtained.



(a) Consider the initial slopes of the curves, and hence state whether T_1 or T_2 is the higher temperature. (1 mark)

(b) Explaining your choice, determine the sign of the enthalpy change for the reaction. (3 marks)

(c) Write the K_c expression for the following reaction: (1 mark)

$$CH_{4(g)} + H_2O_{(g)} \rightleftharpoons 3H_{2(g)} + CO_{(g)}$$

(d) The following concentrations were present in an equilibrium mixture:

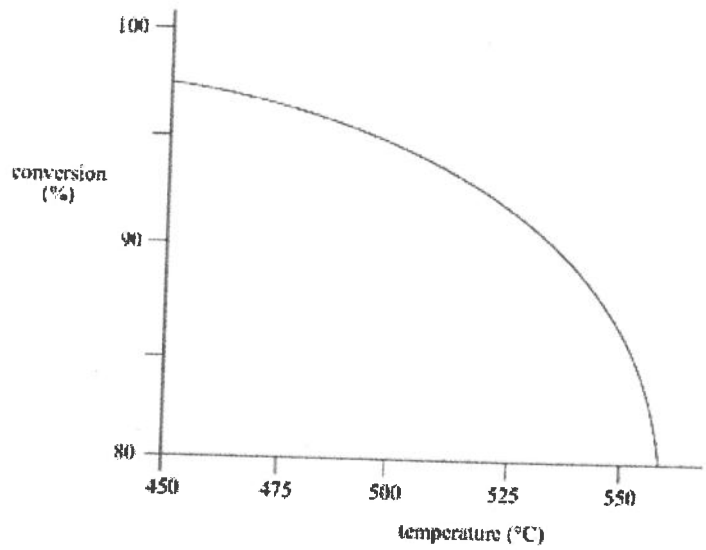
methane	0.25 mol L^{-1}
water	0.10 mol L^{-1}
hydrogen	0.84 mol L^{-1}
carbon monoxide	0.75 mol L^{-1}

Calculate the value of the K_c for this equilibrium mixture. (2 marks)

9. Large quantities of sulfuric acid are required for the treatment of beach sand. Part of the manufacture of sulfuric acid involves the conversion of sulfur dioxide into sulfur trioxide, as shown below:

$$SO_{2(g)} + ½O_{2(g)} \rightleftharpoons SO_{3(g)}$$

The effect of temperature on the percentage conversion of sulfur dioxide into sulfur trioxide is shown in the graph below:



(a) State the effect of increased temperature on the value of K_c. (1 mark)

(b) With reference to the graph, show that the conversion is exothermic. (2 marks)

(c) The temperature for maximum conversion is less than 400°C yet manufacturers tend to operate the plant at 450°C. State a chemical reason for using the higher operating temperature. (1 mark)

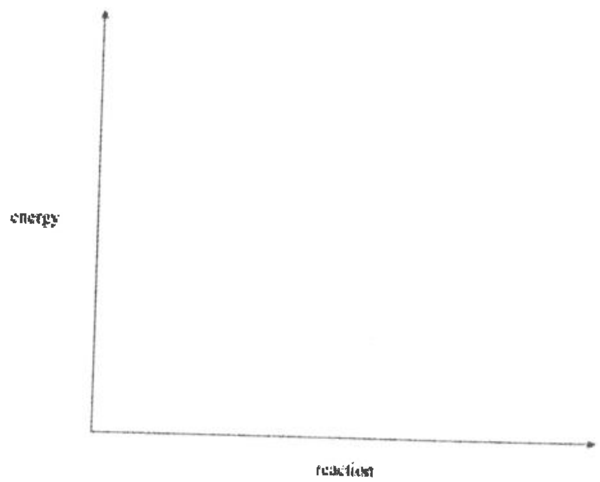
(d) Describe a possible use for the heat produced in this exothermic reaction. (1 mark)

(e) In the conversion of sulfur dioxide into sulfur trioxide, the reacting gases are passed over powdered vanadium pentoxide, which acts as a catalyst.

(i) State why the catalyst is in a powdered form. (1 mark)

(ii) State the effect of the catalyst on the activation energy. (1 mark)

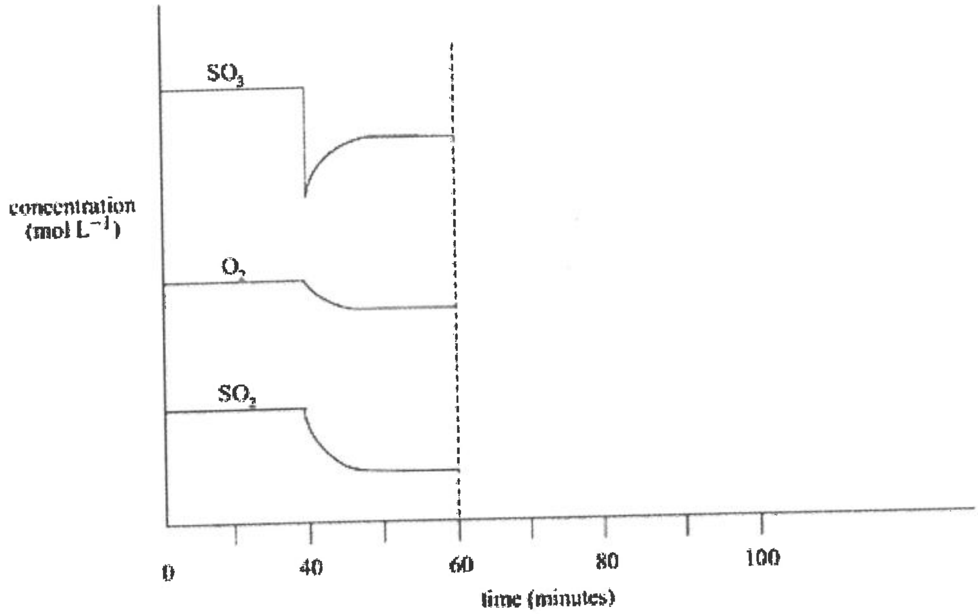
(iii) Complete the following energy profile diagram to show how a catalyst affects the exothermic reaction. (2 marks)



(f) A study of the equilibrium between SO_2, O_2, and SO_3 gases was carried out in a vessel. The equation for the reaction that occurred inside the vessel is shown below:

$$SO_{2(g)} + \frac{1}{2}O_{2(g)} \rightleftharpoons SO_{3(g)} \qquad \Delta H = -96 \text{ kJ}$$

The changes in concentration are shown in the following diagram. Refer to the graphs in answering questions (i) and (ii) below.

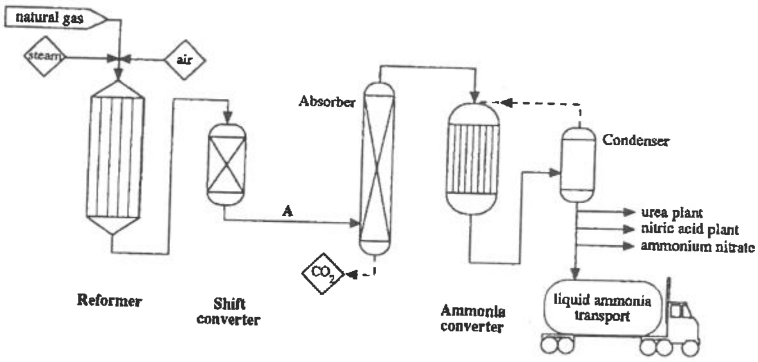


(i) Describe the change in concentration that occurred at the 40-minute mark, and explain the changes that occurred in the reaction mixture during the following 10 minutes. (4 marks)

(ii) At the 60-minute mark the temperature of the vessel was raised. Equilibrium was re- established after 80 minutes. Extend the graph of sulfur trioxide concentration from 60 to 100 minutes. (3 marks)

2.3 Optimising Production

1. Nitrogen may be fixed artificially by industrial processes and then used to produce plant fertilisers. The starting-point for a large number of fertilisers is the production of ammonia. Today most ammonia is produced by means of the Haber process. A simplified diagram of this process is shown in the flow chart below:



The following reactions occur in the vessels:

Reformer: At 750°C and 30 atmospheres using nickel catalyst:

$$CH_{4(g)} + H_2O_{(g)} \rightarrow CO_{(g)} + 3H_{2(g)}$$

$$2H_{2(g)} + O_{2(g)} \rightarrow 2H_2O_{(l)}$$

and some $CO_{(g)} + H_2O_{(g)} \rightarrow CO_{2(g)} + H_{2(g)}$

Shift converter: Carbon monoxide and steam are removed by using iron oxide catalyst at 400°C and copper catalyst at 220°C:

$$CO_{(g)} + H_2O_{(g)} \rightarrow CO_{2(g)} + H_{2(g)}$$

Ammonia converter: At 400°C and 250 atmospheres using iron catalyst:

$$N_{2(g)} + 3H_{2(g)} \rightleftharpoons 2NH_{3(g)} \qquad \Delta H = -92 \text{ kJ}$$

(a) (i) Catalysts are used extensively throughout the process. State and explain their function with the aid of an energy profile graph. (3 marks)

(ii) Write the formulae of three gases that would be present in the pipe marked A in the flow chart. (3 marks)

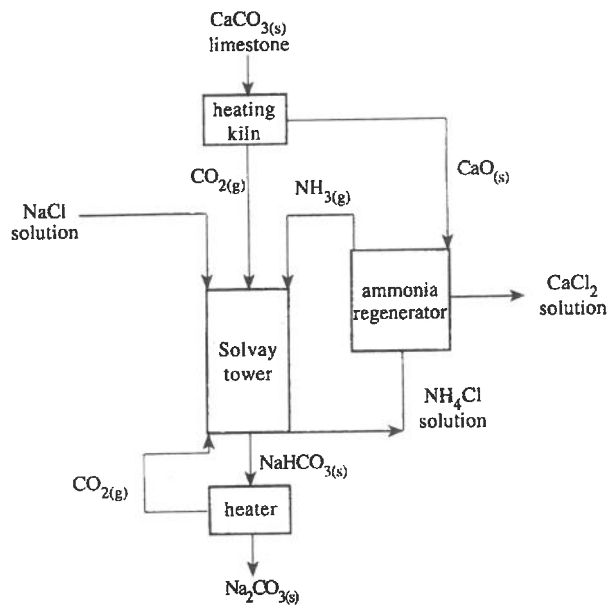
(iii) Suggest one way in which energy costs could be minimised in this process. (1 mark)

(iv) Name the raw materials used in this process. (3 marks)

(v) Name the by-product of this process. (1 mark)

TOPIC 2

2. Sodium carbonate is a chemical produced and used in huge quantities throughout the world. It is produced at an industrial plant by the Solvay process. The key steps in the Solvay process are shown in the flow chart below:

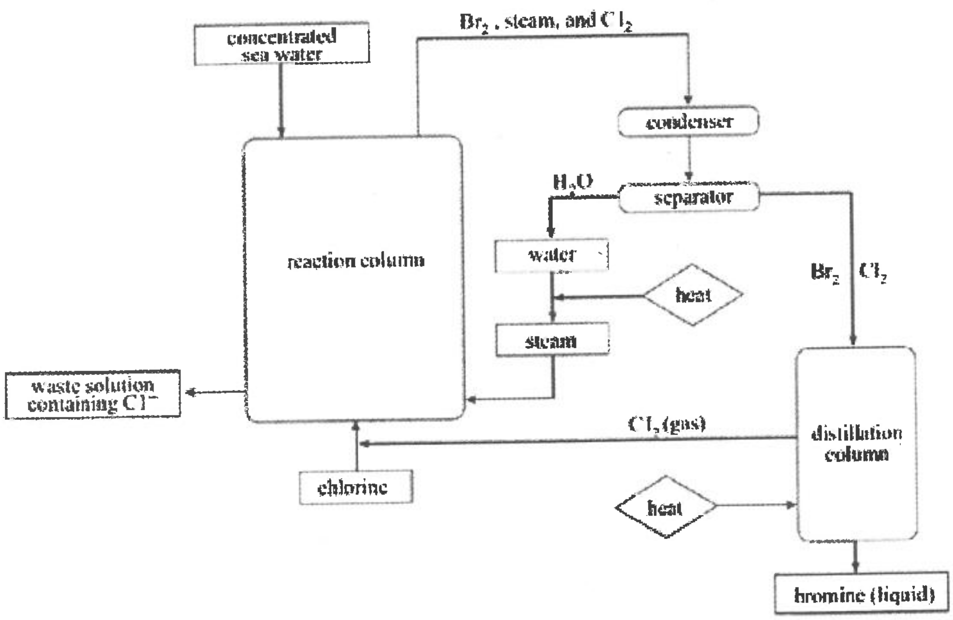


(a) Write an equation for the overall reaction that occurs in the Solvay tower. (2 marks)

(b) Name the solid product that crystallises out in the Solvay tower. (1 mark)

(c) Name a substance that is continually recycled in the Solvay process. (1 mark)

3. The process used for the industrial extraction of bromine is shown in the flow chart below:



(a) Name two raw materials used in the extraction of bromine. (2 marks)

(b) Explain why liquid bromine and water form separate layers in the separator. (2 marks)

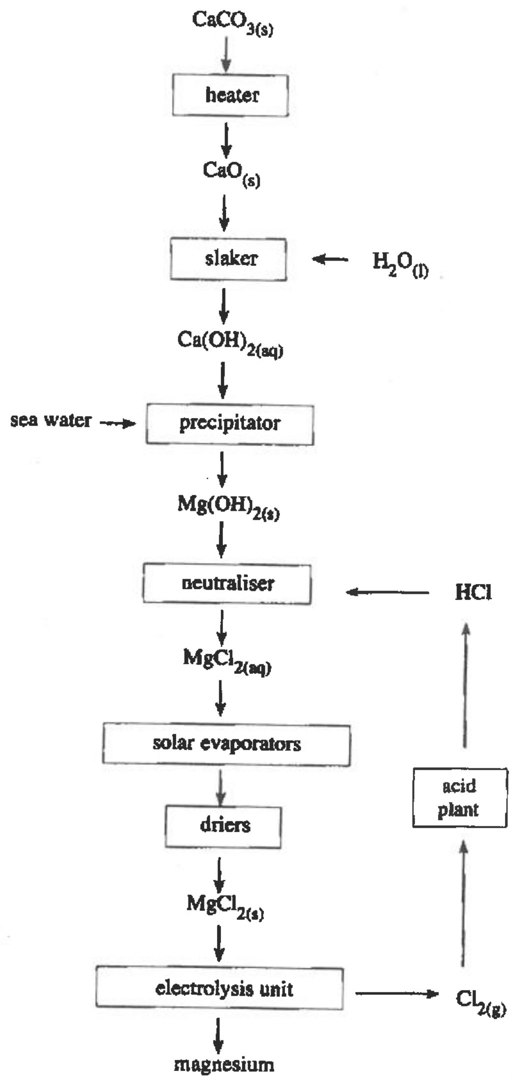
(c) Name one ionic substance that could be extracted from the waste solution. (1 mark)

(d) State one economic and one environmental disadvantage of the production of heat required for multiple stages of the process. (2 marks)

(e) State and explain one way in which the consumption of raw materials is minimised in the production process, thus reducing costs. (3 marks)

(f) Describe one other way in which costs may be reduced from the production of heat in the process. (2 marks)

4. Magnesium is produced from sea water by a series of processes, as shown in the flow chart.



(a) Sea water contains magnesium ions with a concentration of 1.29 g L^{-1}.

(i) Name a technique used for determining the concentration of very small amounts of metal ions in sea water. (1 mark)

(ii) Calculate the volume of sea water that must be processed for 1.00 tonne (1 tonne = 1000 kg) of magnesium to be produced. (2 marks)

(iii) Name the other substance produced when calcium carbonate is converted into calcium oxide. (1 mark)

(iv) From the flow chart, identify *two* stages at which large amounts of energy are needed. (2 marks)

(v) Write the ionic equation for the formation of magnesium hydroxide in the precipitator. (2 marks)

(vi) Explain why all of the water must be removed from the magnesium chloride before it is electrolysed. (2 marks)